# Shaft Supply

# (Mechanical and Electrical Eccentricity for Measurement with Vibration Sensors)

**Revisions**

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## 1 Purpose and scope

This procedure is intended to provide guidelines for the identification and elimination (or reduction to a minimum established value) of errors in the measurement of the vibration amplitude carried out on shafts purchased from Trillium Pumps Italy and Termomeccanica Pompe when the application of proximity transducers is required.

## 2 Acronyms

**SA**: Purchase Specification

**MT**: Magnetoscopic

## 3 Reference documents

### 3.1 Purchase order

The purchase order describes the basic requirements of the supply: material code and description, required material code, quantity, weight, delivery date, destination order and price details.

Any additional requirements in addition to the standard requirements are detailed or referred to on the purchase order.

The checks to be carried out on the shafts are indicated in the purchase order which takes precedence over this SA.

## 4 Supplier's Responsibility

The supplier must issue an Order Confirmation within 10 calendar days of receiving the order, highlighting the acceptance of the contractual requirements expressed in the purchase order, including the delivery dates.

If the order confirmation does not arrive within 10 calendar days, the order will be considered accepted in its entirety, including what is described in this purchase specification.

The supplier is required to request from Trillium Pumps Italy and Termomeccanica Pompe any documents cited in the purchase order and not in its possession and to immediately highlight any discrepancies found between the applicable documents.



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## 5 Supply requirements

### *5.1 Definition of eccentricity*

It may happen that, after installing a vibration transducer, the rotation of the shaft alone causes even significant variations in the signal received by the reading instrument.

These variations, not resulting from actual shaft vibrations, can be attributed to mechanical and electrical causes and are defined as “mechanical and electrical eccentricities”.

### *5.2 Mechanical eccentricity*

It is due to the shaft surface not being perfectly machined on the lathe or by grinding, or to slight deflections of the shaft itself caused by incorrect storage over a long period of time.

These conditions give rise to a frequency signal coinciding with the rotation speed.

### *5.3 Induced electrical eccentricity*

Proximity transducers generally work satisfactorily in the presence of a uniform magnetic field.

If the shaft has areas of non-homogeneous magnetization, a signal variation may occur. This may be due to residual magnetism resulting from mechanical processing or to checks carried out with magnetic particles (MT).

Even microscopic segregations of alloy elements with consequent non-homogeneous metallurgical composition can give signal variations, as well as stress concentrations due to mechanical workings or heat treatments which, even if they do not negatively influence the mechanical properties of the shaft, cause modifications of the surface resistivity.

### *5.4 Test conditions*

Eccentricity measurements must be made on the diameter of the shaft on which the proximity transducer will be installed (indicated in the “Diamond burnishing tool” machining drawings).

Both mechanical run-out measurements must be performed with comparators or contact probes with at least a thousandth precision and electrical run-out measurements must be performed with eddy current proximity probes calibrated for the correct shaft material.

The measuring device must have the following minimum characteristics:



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1. Ensure stable and constant height support of the shaft on the working surfaces of the plain bearings.
2. Do not be subjected to magnetic fields that may alter the proximity probe readings.
3. Ensure smooth and force-free rotation of the shaft without scratching the bearing surfaces (e.g. driven rollers, diabase V-blocks).
4. Both manual benches and the use of automatic benches are permitted.
5. All measuring instruments used must be calibrated within the ranges prescribed by the quality system of the test operator.

The readings taken with the comparator represent the mechanical eccentricity of the shaft.

The readings taken with the transducer represent the total eccentricity of the shaft.

To obtain the electrical eccentricity of the shaft, it is sufficient to algebraically subtract the mechanical eccentricity (measured in the same angular position) from the total eccentricity.

*5.5 Acceptability*

The supplier must comply with the following acceptability criteria:

- The total eccentricity (combined electrical - mechanical run out) must not exceed the maximum between 25% of the maximum permissible peak-to-peak vibration amplitude

  ($A_{max}$) or 6 μm. The vibration amplitude $A_{max}$ is calculated with the following formula:

$$A_{max} = 25.4*(8000/rpm)^{0.5}$$

- Electro-mechanical run-out limits must be met in each section of the proximity probe target areas.
- If the electrical-mechanical runout exceeds the indicated limits, the residual magnetism of the shaft must be checked (measured with a Gaussmeter): each measured section must have a magnetism lower than 2 gauss and have a variation between the maximum and minimum measured value equal to 1 gauss. This check is necessary to determine the type of corrective action to be taken (see paragraph 5.6).



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*5.6 Corrective operations*

If the results obtained are unsatisfactory, the supplier may take action to resolve the shaft non-conformity. (To be communicated to Trillium Pumps Italy and Termomeccanica Pompe).

In particular, the following interventions are useful:

1. Rework the shaft by turning or grinding (mechanical run-out values less than 5 microns are typically required to achieve total electromechanical run-out within specified limits).
2. "Diamond finishing" of the shaft surface where the transducer works.
3. Demagnetizing the shaft using magnetic yokes (repeatedly invert the polarity of the yoke while progressively decreasing the polarization current).

   To allow the correction of electromechanical run out with methods 1 and 2 above, subject to approval by Trillium Pumps Italy and Termomeccanica Pompe, the target diameters of the probes can be reduced (value to be agreed on a case-by-case basis with the technical office of Trillium Pumps Italy and Termomeccanica Pompe); in any case, there must not be steps in the target areas, which in any case would falsify the reading of the proximity probes.

## 6 Protection and transport

It is necessary to refer to SA.05.01.01 "Mechanical processing - full supply and processing account".

The transport of parts must take place in accordance with the Trillium Pumps Italy and Termomeccanica Pompe G.09.08.04 procedure.

## 7 Run out certificate

Preferably 3 days before delivery of the shafts, or at the latest on the same day, the supplier must provide the certificate requested in the Purchase Order in digital format.

The certificate is to be considered an integral part of the supply.

The supplier must send the certificate to the following mailbox:

**Gabbioneta.Certificati@trilliumflow.com**

**IT006.Certificati@trilliumflow.com**



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In order to allow correct traceability between piece and certificate, the casting number and the material code must be reported:

- on the run out certificate
- on the finished piece in the keyway on the joint side in an indelible manner by marking with stamps or pneumatic pen (the code of the marked material must actually correspond to that reported on the order)

The required language for certificates is English, however bilingual Italian-English certificates are acceptable. Any requests for certification in different languages will be communicated.

The certificate must contain the following elements:

1) Indication of order references Trillium Pumps Italy and Termomeccanica Pompe, drawing number and piece number, for complete traceability.
2) Indication of the shaft material.
3) Table of proximity probe measurements in volts, and corresponding values in microns, at least every 10° for tests performed manually and polar diagram of the run out (angular position 0 corresponding to the tab seat, in the drive side hub; in the case of a hub with a double opposing tab, the angular position 0 must correspond to the position of the tab of the first phase impeller).
4) Comparator/contact probe measurement table, at least every 10° for manual tests and run out polar diagram (0° reference as for proximity probes.
5) Electrical run out measurement table.
6) Clear indication of the total run out, only mechanical and only electrical (resulting from the tables requested at points 3, 4 and 5).
7) Indication of the shaft rotation speed and the acceptability limits according to paragraph 5 of this procedure.
8) Calibration certificates of the instruments used.

If the supplier does not have its own format for issuing the certificate, it must request the following facsimiles from Trillium Pumps Italy and Termomeccanica Pompe: Mod.051 Electrical Mechanical run-out certificate and/or Mod.029C Shaft mechanical run-out certificate.



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## 8 Supplier Quality, Health, Safety, Environment, Social Responsibility Requirements

The supplier is required to produce the parts and/or provide services by applying a management system compliant with the requirements of the **ISO 9001 standard** for quality, compliant with the requirements of the **ISO 45001 standard** for Health and Safety at work, compliant with the requirements of the **ISO 14001 standard** for environmental protection, compliant with the requirements of the international standard **SA8000** on corporate social responsibility.

Trillium Pumps Italy and Termomeccanica Pompe reserve the right to check the suitability of the procedures, means and equipment used by the Supplier to ensure and certify the quality of the product in compliance with the environment and the health and safety of workers.

Such checks may take place at any stage of the production process, including visits to the supplier, and may have the purpose of both verifying the product's conformity to specifications and evaluating the supplier's management system.

The results of the above-mentioned checks, the non-conformities detected on the supplies and the compliance with the delivery dates will contribute to the evaluation of the supplier and will be taken into due consideration in the phase of assigning new orders.



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